Here's all topics:

Data Sampling & Reference

- Population & Sample
- Reference & Sampling Technique

Types of Data

- **Qualitative (Categorical) Data**
  - Nominal
  - Ordinal
- **Quantitative (Numerical) Data**
  - Discrete
  - Continuous

Data Collection & Representation

- Cross Sectional Data & Time Series Data
- Relative Frequency, Cumulative Frequency
- Histogram
- Scatter Plots
- Box Plot, Outliers

Types of Statistics

Descriptive Statistics

- **Measures of Central Tendency:**
  - Mean, Mode & Median (Their frequency distribution)
- **Measures of Dispersion:**
  - Range
  - Variance
  - Standard Deviation
  - Interquartile Range (IQR)
  - Mean Absolute Deviation (MAD)
  - Coefficient of Variation
  - Covariance

Data Distribution

- Skewness (Positive Skew, Negative Skew, Zero Skew)
- Kurtosis (Leptokurtic, Mesokurtic, Platykurtic)

Types of Variables

- **Categorical Variables:**
  - Nominal Variable
  - Ordinal Variable
- **Numerical Variables:**
  - Discrete
  - Continuous
- Dependent Variable
- Independent Variable
- Control, Moderating & Mediating Variables

Data Table Construction

- Range
- Calculate Class Width
- Create Intervals
- Count Frequencies
- Construct the Table

Statistical Analysis & Correlation

- Pearson & Spearman Correlation Methods
- Regression Error Metrics
- Percentiles, Quartiles, Inner Quartile Range
- Different Types of Plots for Continuous & Categorical Variables

Statistical Inference & Hypothesis Testing

- Confidence Intervals
- Central Limit Theorem
- Degree of Freedom

Loss Functions

- Mean Squared Error (MSE), Mean Absolute Error (MAE) – Loss Function
- Huber Loss Function
- Cross Entropy Loss Function

Hypothesis Testing

- One-tail, Two-tail & p-value
- Formulation of Null & Alternate Hypothesis

- Type-I Error & Type-II Error
- Sample Test
- ANOVA Test
- Chi-square Test
- Z-Test & T-Test

What is Statistics?

Statistics is the study of data. It helps us collect, organize, analyze, and understand data to make better decisions. In data science, statistics is used to find patterns, make predictions, and check if results are reliable. It is important for machine learning, business decisions, healthcare, and many other areas.

Example: Favorite Ice Cream Survey

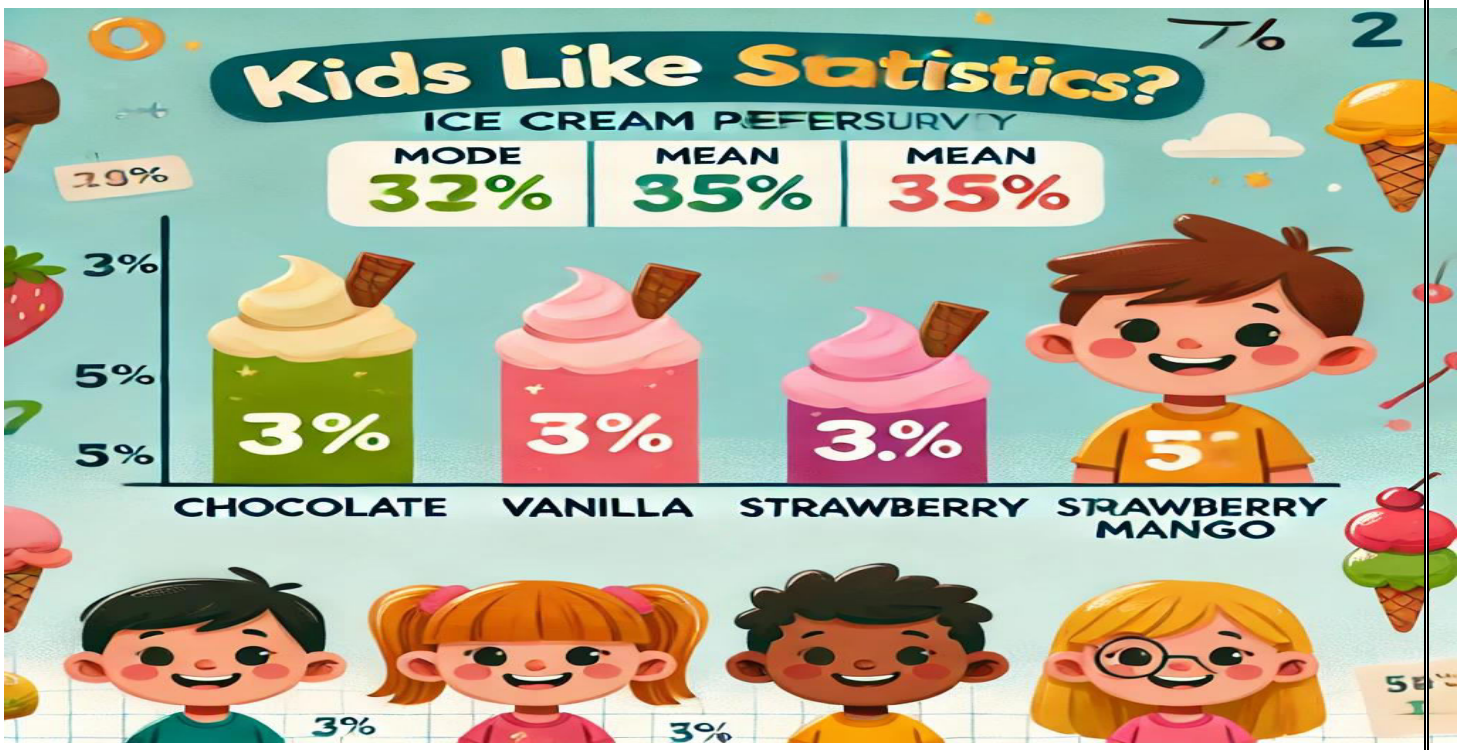
Imagine we ask 10 kids about their favorite ice cream flavors. Here's what they say:

- 4 like **Chocolate**
- 3 like **Vanilla**
- 2 like **Strawberry**

- 1 likes **Mango**

How Statistics Helps Us Understand This Data:

1. **Mode** – The most popular flavor.
    - **Chocolate** is the mode because most kids chose it.
2. **Mean (Average)** – If we assigned numbers (Chocolate = 4, Vanilla = 3, etc.), we could find the average preference.
3. **Bar Chart** – A simple visual to show how many kids like each flavor.

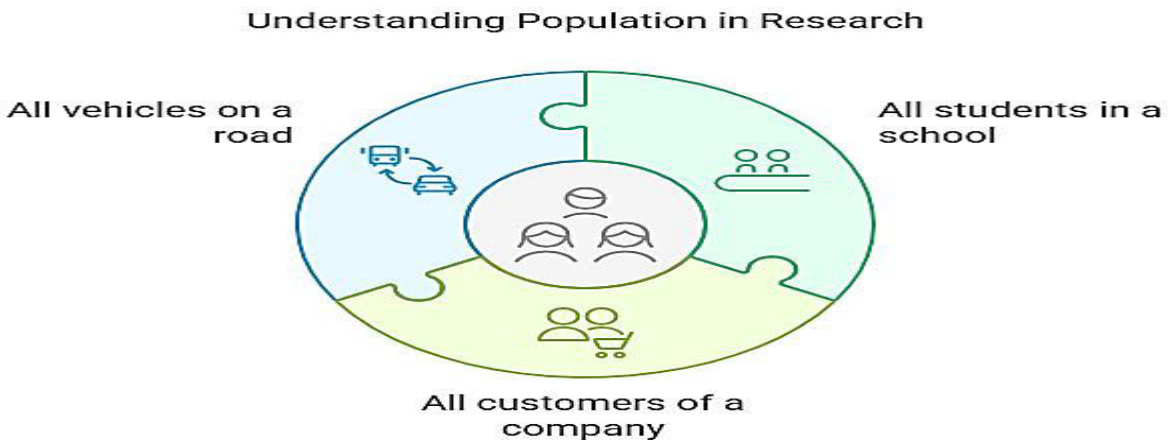


Data Sampling & Reference

1. Population & Sample

Population:

- The entire group of individuals, items, or data points that we want to study.
- Example: All students in a school, all customers of a company, or all vehicles on a road.



Sample:

- A subset of the population selected for analysis.
- Used when studying the entire population is impractical.
- Example: A group of 100 students chosen from a school of 1,000 students.

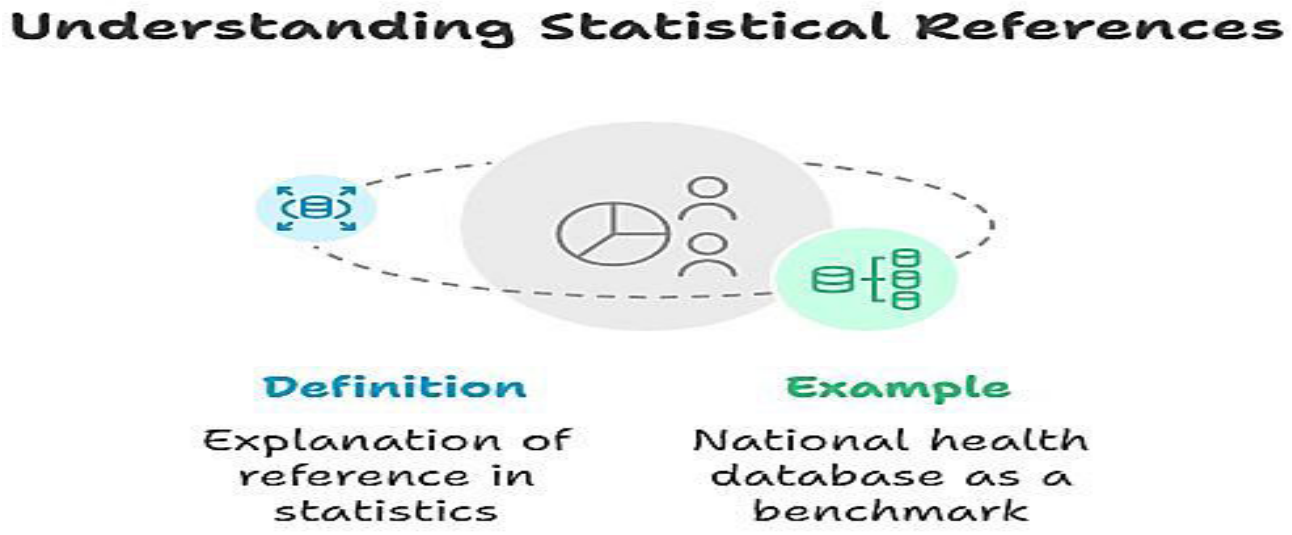
Key Differences:

Feature	Population	Sample
Size	Large	Small
Cost	High	Low
Time Required	More	Less
Accuracy	High	Can be Approximate

2. Reference & Sampling Technique

Reference:

- In statistics, a reference refers to the source or standard dataset against which a sample is compared.
- Example: A national health database can serve as a reference for a local health survey.



Sampling Techniques:

1. Probability Sampling (Random Sampling)

- Every element in the population has an equal chance of being selected.
- Ensures unbiased representation.
- Types:
  - **Simple Random Sampling**: Each individual has an equal probability of selection.
  - **Stratified Sampling**: Population divided into subgroups, then random sampling is performed within each subgroup.
  - **Cluster Sampling**: Population is divided into clusters, and entire clusters are randomly selected.
  - **Systematic Sampling**: Selecting every nth individual from a population list.

2. Non-Probability Sampling (Non-Random Sampling)

- Not every individual has an equal chance of being selected.
- Used when random sampling is difficult or not necessary.
- Types:
  - **Convenience Sampling**: Selecting individuals that are easiest to access.
  - **Judgmental (Purposive) Sampling**: Selection based on researcher's judgment.
  - **Snowball Sampling**: Existing participants recruit new participants.
  - **Quota Sampling**: Selecting a specific quota of individuals based on certain characteristics.

Choosing the Right Sampling Technique:

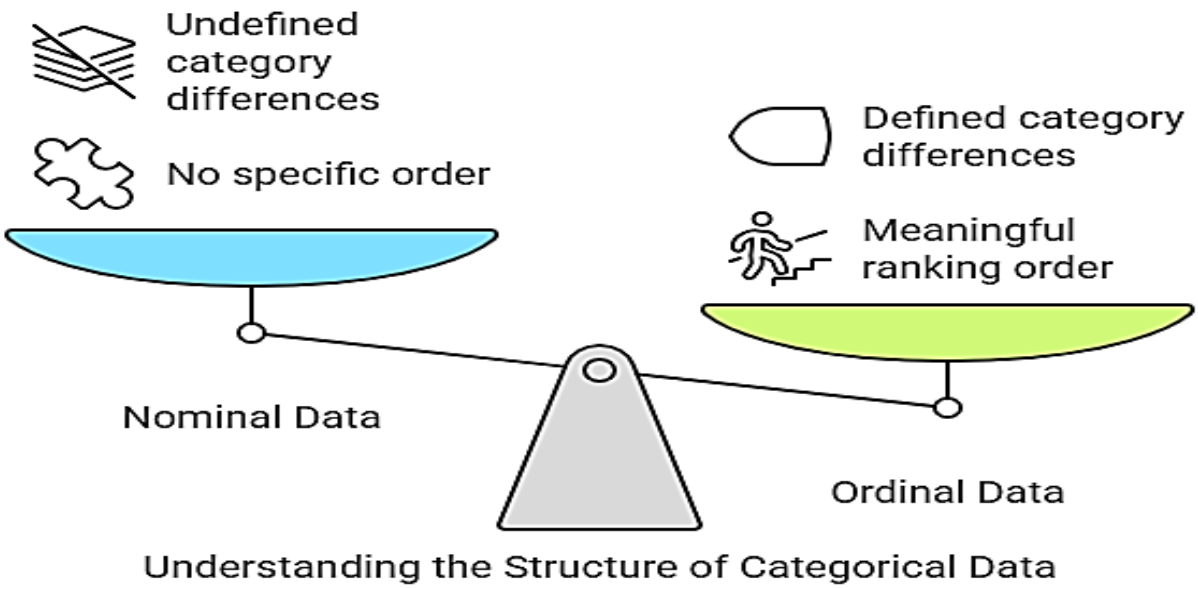
Factor	Probability Sampling	Non-Probability Sampling

Accuracy	High	Can be biased
Time	More time required	Less time required
Cost	High	Low
Suitability	Large, diverse populations	Small, targeted populations

Types of Data

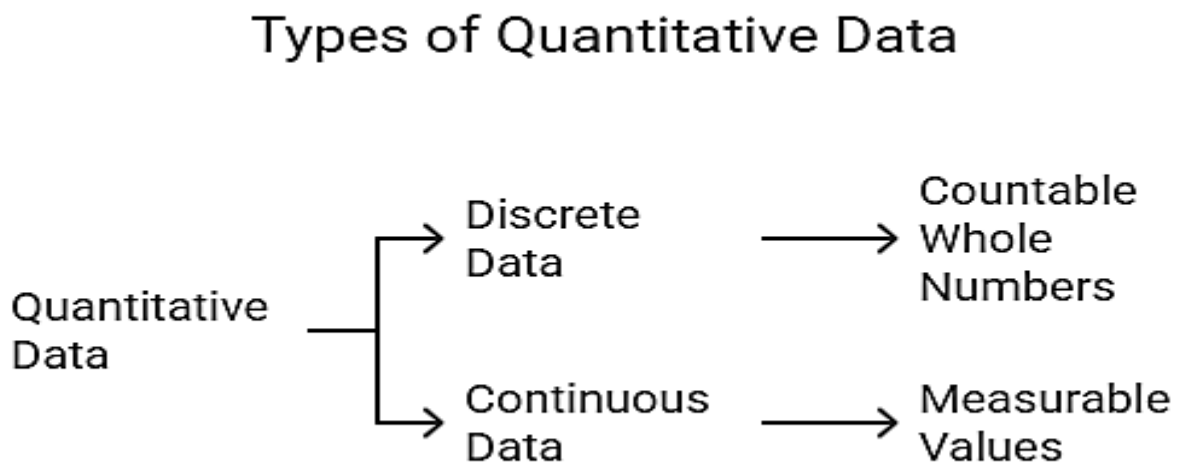
1. Qualitative (Categorical) Data

- **Nominal** – Categories without a specific order (e.g., gender, product type, city names).
- **Ordinal** – Ordered categories with meaningful ranking but undefined differences (e.g., movie ratings, education levels).



2. Quantitative (Numerical) Data

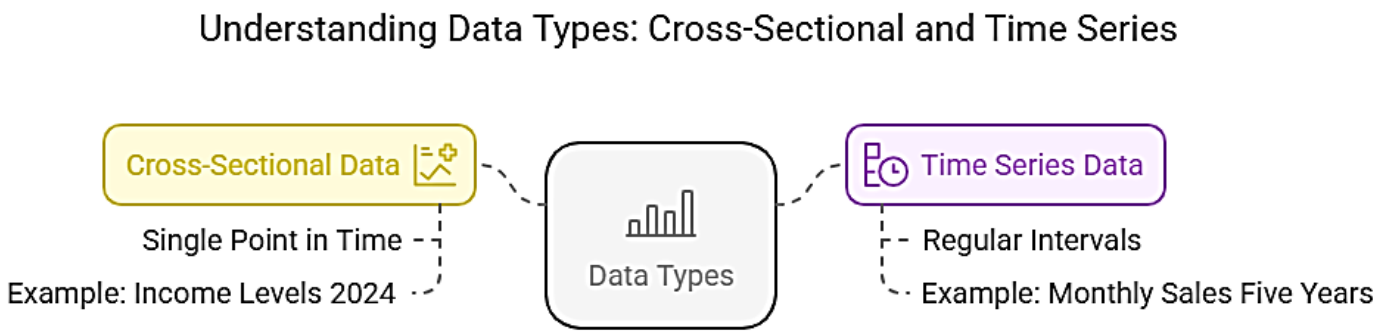
- **Discrete** – Countable whole numbers (e.g., number of website clicks, sold items).
- **Continuous** – Measurable values with infinite possibilities (e.g., temperature, stock prices, customer age).



Data Collection & Representation

1. Cross-Sectional Data & Time Series Data

- **Cross-Sectional Data** – Data collected at a single point in time (e.g., income levels of households in 2024).
- **Time Series Data** – Data collected over time at regular intervals (e.g., monthly sales of a company over five years).



2. Relative Frequency & Cumulative Frequency

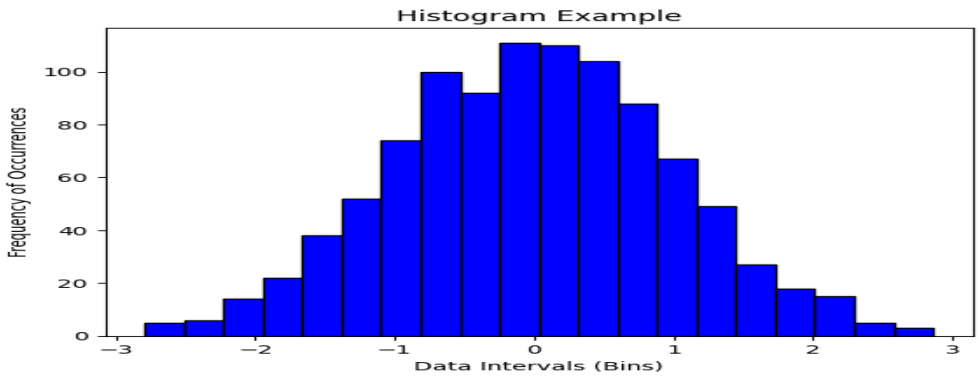
- **Relative Frequency:** The proportion of occurrences of a specific value compared to the total observations.

$$\text{Relative Frequency} = \frac{\text{Frequency of a value}}{\text{Total observations}}$$

- **Cumulative Frequency:** A running total of frequencies up to a certain value.

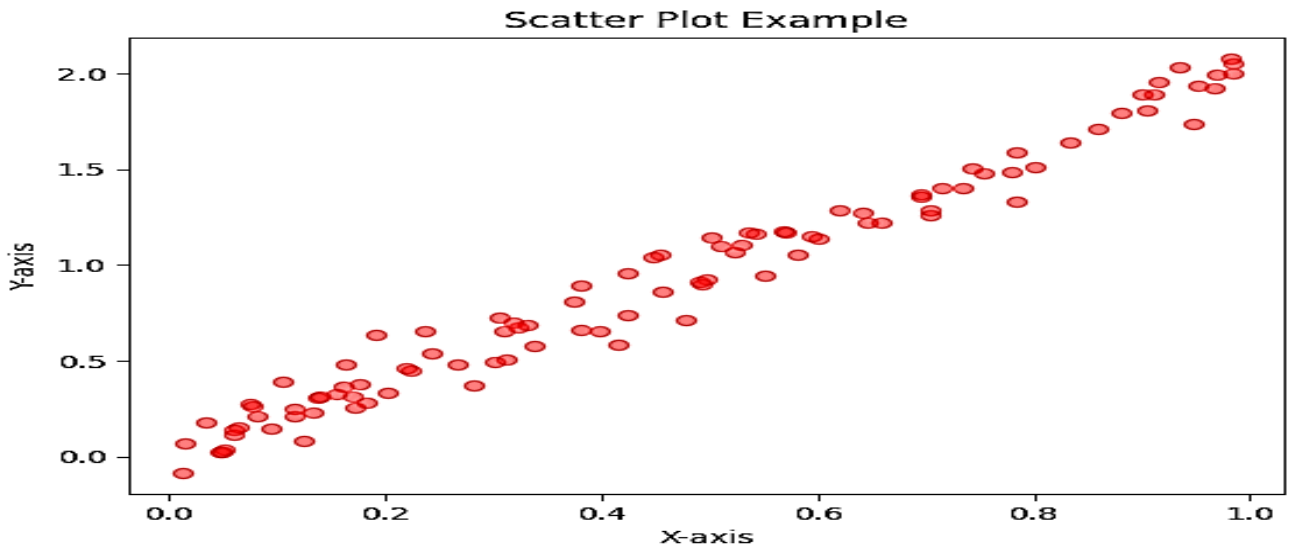
3. Histogram

- A **bar chart** representing the frequency distribution of numerical data.
- X-axis: Data intervals (bins).
- Y-axis: Frequency of occurrences..



4. Scatter Plots

- A graph used to show the relationship between two numerical variables..
- **Correlation types**:
- Positive: Both variables increase together.
- Negative: One variable increases while the other decreases.
- No correlation: No apparent relationship.



5. Box Plot & Outliers

Box Plot: A graphical representation of a dataset's **five-number summary**:

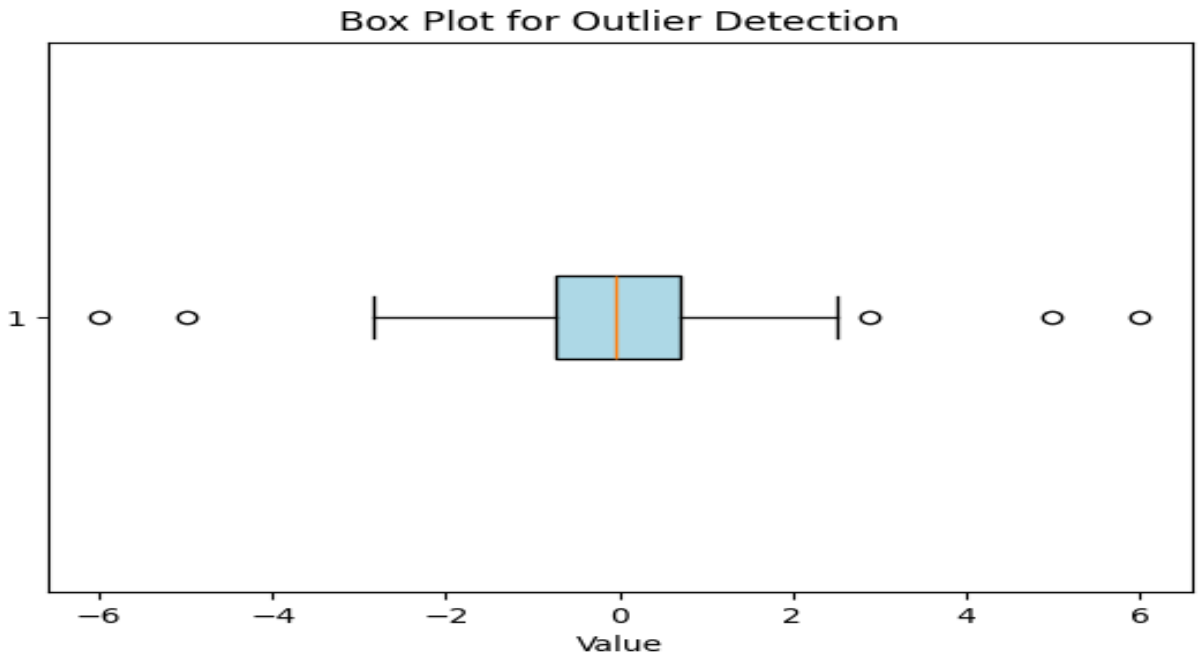
- Minimum
- First Quartile (Q1)
- Median (Q2)
- Third Quartile (Q3)
- Maximum

- **Outliers:** Extreme values outside the normal range, often detected using the **Interquartile Range (IQR):**

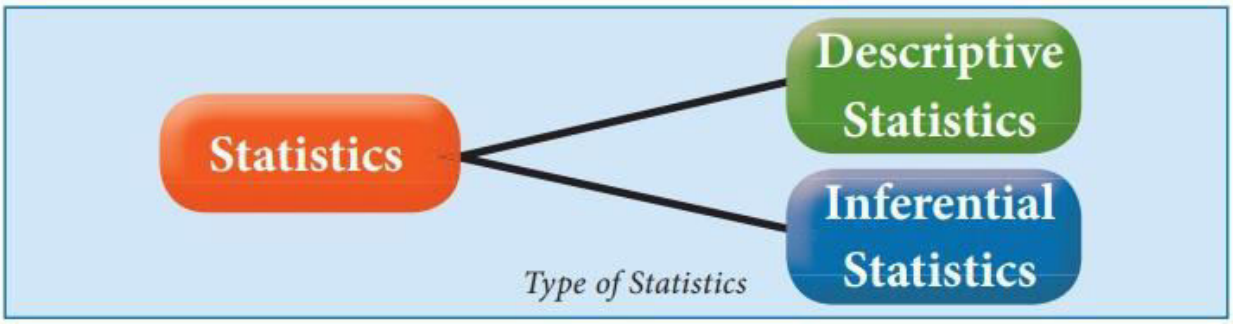
$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

where $IQR = Q3 - Q1$.



Types of Statistics



1. **Descriptive Statistics**:
   - **Definition**: Descriptive statistics summarize and describe the main features of a dataset. It provides simple summaries about the sample and the measures.
   - **Key Concepts**:
     - **Measures of Central Tendency**: Mean, Median, Mode.
     - **Measures of Dispersion**: Range, Variance, Standard Deviation.
     - **Frequency Distribution**: Histograms, Bar Charts.
     - **Percentiles and Quartiles**: Interquartile Range (IQR).
   - **Example**: Calculating the average income of a group of people or the standard deviation of test scores.
2. **Inferential Statistics**:
   - **Definition**: Inferential statistics use a random sample of data taken from a population to make inferences or predictions about the population. It helps in testing hypotheses and making generalizations.

- **Key Concepts**:
  - **Hypothesis Testing**: Null and Alternative Hypotheses, p-values, Significance Levels.
  - **Confidence Intervals**: Estimating population parameters.
  - **Regression Analysis**: Linear and Logistic Regression.
  - **ANOVA (Analysis of Variance)**: Comparing means across groups.
- **Example**: Predicting the average height of all students in a university based on a sample of 100 students.

Key Differences Between Descriptive and Inferential Statistics

Aspect	Descriptive Statistics	Inferential Statistics
Purpose	Summarize and describe data	Make predictions or inferences about a population
Data Used	Entire dataset	Sample data from the population
Techniques	Mean, Median, Mode, Standard Deviation, Histograms	Hypothesis Testing, Confidence Intervals, ANOVA
Example	Calculating the average age of a class	Predicting the average age of all students in a school based on a sample

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No.	Student	Age
1	Arya	28
2	Mohit	28
3	Pawan	30
4	Suresh	32
5	Kalyan	34
6	Sagar	35
7	Maya	36

Data Distribution

Data distribution describes how data points are spread across a dataset. Two key characteristics of distribution are **Skewness** and **Kurtosis**.

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1. Skewness

Skewness measures the **asymmetry** of the data distribution.

Types of Skewness:

1. **Positive Skew (Right-Skewed)**
   - **Tail is longer on the right**
   - **Mean > Median > Mode**
   - Example: Income distribution (few very high incomes pull the mean up)
2. **Negative Skew (Left-Skewed)**
   - **Tail is longer on the left**
   - **Mean < Median < Mode**
   - Example: Exam scores (most students score high, a few score very low)
3. **Zero Skew (Symmetric Distribution)**
   - **Mean ≈ Median ≈ Mode**
   - Example: Normal distribution (e.g., IQ scores)

Formula for Skewness:

$$Skewness = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- $\bar{x}$ = Mean
- s = Standard deviation
- n = Number of observations

2. Kurtosis

Kurtosis measures the **tailedness** (how extreme the outliers are) of a distribution.

Types of Kurtosis:

1. **Leptokurtic (Kurtosis > 3)**
   - **Heavy tails**, more outliers
   - Example: Stock market returns

2. **Mesokurtic (Kurtosis ≈ 3)**
    - **Normal distribution** (moderate tails)
    - Example: Heights of people
3. **Platykurtic (Kurtosis < 3)**
    - **Light tails**, fewer extreme values
    - Example: Uniform distribution

Formula for Kurtosis:

$$Kurtosis = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

Types of Variables in Data Science

Variables are classified into different types based on their nature and role in statistical analysis and machine learning.

1. Categorical Variables (Qualitative)

Categorical variables represent groups or categories. They **do not have numerical meaning**.

a) Nominal Variables

- Categories with **no natural order** or ranking.
- Example:
    - Gender (Male, Female, Other)
    - Colors (Red, Blue, Green)
    - Car Brands (Toyota, Ford, BMW)

b) Ordinal Variables

- Categories with a **meaningful order**, but differences between them **are not measurable**.
- Example:
    - Education Level (High School < Bachelor's < Master's < PhD)
    - Customer Satisfaction (Poor < Average < Good < Excellent)

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2. Numerical Variables (Quantitative)

Numerical variables represent measurable quantities.

a) Discrete Variables

- **Whole numbers** (countable values).
- Example:

- Number of children in a family (0, 1, 2, …)
- Number of cars in a parking lot (5, 10, 15)

b) Continuous Variables

- Can take **any value within a range** (including decimals).
- Example:
  - Height (170.5 cm, 175.2 cm)
  - Temperature (36.6°C, 98.4°F)
  - Distance (5.25 km, 10.75 km)

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3. Dependent vs. Independent Variables

Used in predictive modeling and experimental research.

a) Dependent Variable (Target, Response Variable)

- The **outcome** we want to predict or measure.
- **Depends on** other variables.
- Example:
  - Exam Score (dependent on study hours)
  - Sales Revenue (dependent on advertising spend)

b) Independent Variable (Predictor, Explanatory Variable)

- The **cause** or factor influencing the dependent variable.
- Example:
  - Study Hours (affecting exam score)
  - Advertising Spend (affecting sales revenue)

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4. Other Important Variables

a) Control Variable

- Kept **constant** to avoid influencing the results.
- Example:
  - Studying the effect of caffeine on productivity → **Sleep duration** is a control variable.

b) Moderating Variable

- **Affects the strength or direction** of the relationship between independent and dependent variables.
- Example:
  - The impact of exercise on weight loss → **Diet** is a moderating variable.

c) Mediating Variable

- **Explains the relationship** between independent and dependent variables.
- Example:
  - Higher education → **Job skills (Mediator)** → Higher salary.

Data Table Construction

Data Table Construction Step-by-Step

Constructing a data table involves organizing raw data into a structured format, typically in a frequency distribution table.

1. Range

The **range** is the difference between the **maximum** and **minimum** values in the dataset.

$$\text{Range} = \text{Max Value} - \text{Min Value}$$

Example:

Data: **[5, 8, 12, 15, 18, 21, 24, 30]**

- Max Value = 30
- Min Value = 5
- **Range = 30 - 5 = 25**

2. Calculate Class Width

Class width determines the size of each interval.

$$\text{Class Width} = \frac{\text{Range}}{\text{Number of Classes}}$$

Rules for choosing class width:

- It should cover the entire data range.
- Use **rounding up** for better readability.

Example Calculation:

- Suppose we want **5 classes.**
- **Class Width = 25 / 5 = 5** (rounded up if necessary).

3. Create Intervals

Using the class width, create **non-overlapping** intervals.

Example Intervals (Starting from Min = 5, Class Width = 5):

1. **5 - 9**
2. **10 - 14**
3. **15 - 19**
4. **20 - 24**
5. **25 - 29**

4. Count Frequencies

Count how many data points fall into each interval.

Interval	Frequency
5 - 9	2
10 - 14	1
15 - 19	2
20 - 24	2
25 - 29	0
30 - 34	1

5. Construct the Table

Now, put everything together in a **Frequency Distribution Table**.

Interval	Frequency	Cumulative Frequency
5 - 9	2	2
10 - 14	1	3
15 - 19	2	5
20 - 24	2	7
25 - 29	0	7
30 - 34	1	8

Statistical Analysis & Correlation

1. Pearson & Spearman Correlation Methods

Correlation measures the **relationship** between two variables.

a) Pearson Correlation (Linear Relationship)

- Measures the **strength & direction** of a **linear** relationship.
- Values range from **-1 to +1**:
  - **+1** → Perfect **positive** correlation
  - **0** → No correlation
  - **-1** → Perfect **negative** correlation
- Formula:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

- Assumes **normally distributed data** and **linear relationship.**

b) Spearman Correlation (Rank-Based)

- Measures the **monotonic** relationship (does not require **linearity**).
- Uses **ranks** instead of actual values.
- Formula:

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

  where d_i is the difference between the ranks of corresponding values.
- **Best for ordinal data or non-linear relationships.**

2. Regression Error Metrics

Used to **evaluate regression models**.

a) Mean Absolute Error (MAE)

- Measures the **absolute difference** between actual and predicted values.
- Formula:

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

b) Mean Squared Error (MSE)

- Measures the **squared difference** between actual and predicted values.
- Formula:

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

c) Root Mean Squared Error (RMSE)

- Square root of MSE, interpretable in the original units of the dependent variable.
- Formula:

$$RMSE = \sqrt{MSE}$$

d) R-Squared (R^2)

- Measures **how well the independent variables explain** the dependent variable.
- Formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

  - SS_{res} = Residual sum of squares
  - SS_{tot} = Total sum of squares
- R^2 close to **1** → Good model
- R^2 close to **0** → Poor model

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3. Percentiles, Quartiles & Interquartile Range (IQR)

Used to understand **data spread**.

a) Percentiles

- The **p-th percentile** is the value below which p%p\%p% of the data falls.
- Example: The **90th percentile** means **90% of values are below it**.

b) Quartiles

- Special percentiles dividing data into **4 equal parts**:
  - **Q1 (25th percentile)** → Lower Quartile
  - **Q2 (50th percentile)** → Median
  - **Q3 (75th percentile)** → Upper Quartile

c) Interquartile Range (IQR)

- Measures data **spread** (range of the middle 50%).
- Formula:

$$IQR = Q3 - Q1$$

- Used for **outlier detection**:
  - **Lower Bound** $= Q1 - 1.5 \times IQR$
  - **Upper Bound** $= Q3 + 1.5 \times IQR$

4. Different Types of Plots for Continuous & Categorical Variables

a) Continuous Variables (Numerical Data)

- **Histogram** → Distribution of a single variable
- **Box Plot** → Identifies **outliers** & quartiles
- **Scatter Plot** → Relationship between two continuous variables
- **Line Plot** → Trends over time (useful for time series)

b) Categorical Variables

- **Bar Chart** → Comparison of categories
- **Pie Chart** → Proportional distribution
- **Count Plot** → Frequency of categories
- **Stacked Bar Chart** → Comparison within groups

Statistical Inference & Hypothesis Testing

Statistical inference allows us to draw conclusions about a **population** based on a **sample**. It relies on key concepts like **confidence intervals, the central limit theorem, and degrees of freedom**.

1. Confidence Intervals (CI)

A **Confidence Interval (CI)** provides a range of values within which we expect the true population parameter to lie, with a given level of confidence (e.g., 95%).

Formula for Confidence Interval (CI) for the Mean

$$CI = \bar{x} \pm Z \times \frac{s}{\sqrt{n}}$$

where:

- $\bar{x}$ = Sample mean
- Z = Z-score for confidence level (e.g., 1.96 for 95% CI)
- s = Sample standard deviation
- n = Sample size

Example: 95% Confidence Interval

- Sample Mean = 50
- Standard Deviation = 10
- Sample Size = 100
- Z-score for 95% confidence = 1.96

$$CI = 50 \pm (1.96 \times \frac{10}{\sqrt{100}})$$

$$CI = 50 \pm 1.96$$

$$CI = (48.04, 51.96)$$

Interpretation: We are **95% confident** that the true population mean lies between **48.04 and 51.96**.

2. Central Limit Theorem (CLT)

The **Central Limit Theorem (CLT)** states that, for a **large enough sample size**, the **sampling distribution of the sample mean** will be **approximately normal**, regardless of the shape of the original population distribution.

Key Takeaways

- If the sample size n is large ($n \geq 30$), the **sample mean follows a normal distribution.**
- The **larger the sample size**, the **closer** the sampling distribution is to normal.
- The **standard deviation of the sample mean** is called the **standard error (SE)**:

$$SE = \frac{\sigma}{\sqrt{n}}$$

where σ is the population standard deviation.

Why is CLT Important?

- Enables us to apply **parametric tests** (like hypothesis testing) even if the original data is **not normally distributed**.
- Helps estimate **population parameters** from **sample statistics**.

3. Degrees of Freedom (df)

The **degrees of freedom (df)** represent the number of **independent values** in a dataset that can vary while calculating a statistic.

Formula for Degrees of Freedom

For a **sample of size** n:

$$df = n - 1$$

Why is it Important?

- Used in **t-tests, chi-square tests,** and other hypothesis tests.
- Helps **adjust for variability** in smaller samples.

Example: t-Distribution & df

If we calculate the sample **variance**:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

We use $n - 1$ **instead of** n because **one value is already determined by the mean.**

Loss Functions in Machine Learning

Loss functions measure how well a model's predictions match the actual values.The goal is to **minimize** the loss to improve the model's performance.

1. Mean Squared Error (MSE) – Regression Loss

MSE measures the **average squared difference** between actual and predicted values.

Formula

$$MSE = \frac{1}{n}\sum(y_i - \hat{y}_i)^2$$

where:

- y_i = Actual value
- $\hat{y}_i$ = Predicted value
- n = Number of samples

2. Mean Absolute Error (MAE) – Regression Loss

MAE measures the **average absolute difference** between actual and predicted values.

Formula

$$MAE = \frac{1}{n}\sum|y_i - \hat{y}_i|$$

3. Huber Loss Function – Robust Regression Loss

Huber Loss is a **combination of MSE and MAE**, making it **less sensitive to outliers** while maintaining **differentiability.**

Formula

$$L_\delta(a) = \begin{cases} \frac{1}{2}a^2 & \text{for } |a| \le \delta \quad \text{(MSE-like region)} \\ \delta(|a| - \frac{1}{2}\delta) & \text{for } |a| > \delta \quad \text{(MAE-like region)} \end{cases}$$

where $a = y_i - \hat{y}_i$ and δ is a threshold parameter.

4. Cross-Entropy Loss Function – Classification Loss

Used for **classification tasks**, especially **logistic regression and neural networks**.

Binary Cross-Entropy (for two classes)

$$L = -\frac{1}{n} \sum \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]$$

Categorical Cross-Entropy (for multiple classes)

$$L = -\sum_{i=1}^{n} \sum_{j=1}^{k} y_{ij} \log(\hat{y}_{ij})$$

where:

- y_{ij} = 1 if class j is the correct class, else 0.
- $\hat{y}_{ij}$ = Predicted probability for class j.

Hypothesis Testing in Statistics

Hypothesis testing is a statistical method used to **infer conclusions** about a **population** based on a **sample**. It helps determine whether an observed effect is **statistically significant**.

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1. One-Tail, Two-Tail & p-Value

a) One-Tailed Test

- Used when we expect a **specific direction** of effect (greater or less than).
- **Example:** Testing if a **new drug is better** than the existing one.
- Hypothesis:
  - **H_0** (Null): Drug has **no effect**.
  - **H_1** (Alternate): Drug **improves** health.
- If the **p-value** is **small** (e.g., < 0.05), we reject H_0.

b) Two-Tailed Test

- Used when we check for **any difference** (increase or decrease).
- **Example:** Testing if **average height** of students has **changed**.
- Hypothesis:
  - **H_0**: No change in height.
  - **H_1**: Height has changed (could be **higher or lower**).

c) p-Value (Probability Value)

- Measures the probability of obtaining observed results **if the null hypothesis (H_0) were true**.
- If **$p < 0.05$**, reject H_0 (**statistically significant result**).
- If **$p > 0.05$, fail to reject H_0** (not enough evidence).

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2. Formulation of Null & Alternate Hypothesis

- **Null Hypothesis (H_0)** → Assumes **no effect or no difference**.
- **Alternate Hypothesis (H_1)** → Assumes **a significant effect or difference**.

Example: Testing if a new **marketing strategy** increases sales.

- **H_0:** The strategy **does not** increase sales.
- **H_1:** The strategy **increases** sales.

---

3. Type-I & Type-II Errors

Error Type	Definition	Consequence
Type-I Error (False Positive)	Rejecting H_0 when it is actually true.	Incorrectly concluding an effect exists.
Type-II Error (False Negative)	Failing to reject H_0 when it is actually false.	Missing a real effect.

Example:

- **Type-I Error:** A medical test **incorrectly** detects cancer in a healthy person.
- **Type-II Error:** A test **fails to detect** cancer in a patient who actually has it.

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4. Sample Test

A **sample test** checks if sample data is representative of the population.

- **Z-Test**: Used for **large samples ($n > 30$)** with known population variance.
- **T-Test**: Used for **small samples ($n < 30$)** or unknown variance.

---

5. ANOVA (Analysis of Variance) Test

Used to compare **more than two groups** to see if at least one is significantly different.

Example:

- Comparing exam scores of students across **three different schools.**
- **H_0:** No difference in average scores.
- **H_1:** At least one school has a different average score.

Formula for F-Statistic:

$$F = \frac{\text{Between-group variance}}{\text{Within-group variance}}$$

If **$p < 0.05$**, at least one group **differs significantly.**

6. Chi-Square Test (χ^2 Test)

Used to test relationships **between categorical variables**.

Example:

- Checking if **gender** affects **preference for a new product**.
- **H_0:** Gender and preference are **independent**.
- **H_1:** Gender and preference are **related**.

If **χ^2 statistic** is high and **$p < 0.05$**, reject H_0 (**significant relationship**).

7. Z-Test vs. T-Test

Test	When to Use?	Formula
Z-Test	Large samples ($n > 30$), known variance	$\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$
T-Test	Small samples ($n < 30$), unknown variance	$\frac{\bar{x}-\mu}{s/\sqrt{n}}$

Applications of Statistics

1. **Business and Economics**:

   - Market research, sales forecasting, risk analysis, and financial modeling.
   - Example: Analyzing customer purchase patterns to predict future sales.
2. **Healthcare**:
   - Clinical trials, epidemiology, and public health studies.
   - Example: Determining the effectiveness of a new drug by analyzing patient data.
3. **Social Sciences**:
   - Survey analysis, demographic studies, and behavioral research.
   - Example: Studying the impact of education on income levels.
4. **Machine Learning and Data Science**:
   - Feature selection, model evaluation, and data preprocessing.
   - Example: Using regression analysis to predict house prices based on features like location and size.

Common Statistical Tools and Software

1. **Excel**: Basic statistical analysis and visualization.
2. **R**: A programming language specifically designed for statistical computing.
3. **Python**: Libraries like Pandas, NumPy, SciPy, and StatsModels for statistical analysis.
4. **SPSS**: A software package used for statistical analysis in social sciences.
5. **SAS**: Advanced analytics, multivariate analysis, and predictive modeling.

Key Statistical Concepts

1. **Probability**:
   - The likelihood of an event occurring, ranging from 0 (impossible) to 1 (certain).
   - Example: The probability of rolling a 6 on a fair die is 1/6.
2. **Distributions**:
   - **Normal Distribution**: Bell-shaped curve, symmetric around the mean.
   - **Binomial Distribution**: Probability of success/failure in a fixed number of trials.

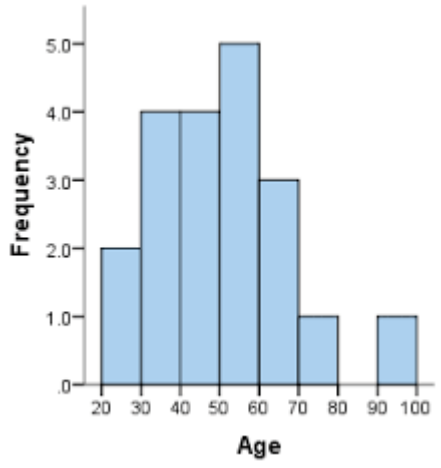
- **Poisson Distribution**: Probability of a given number of events happening in a fixed interval.

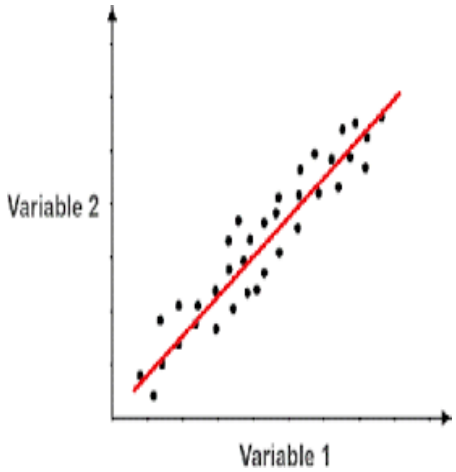
3. **Correlation vs. Causation**:
   - **Correlation**: A statistical relationship between two variables.
   - **Causation**: One variable directly affects the other.
   - Example: Ice cream sales and drowning incidents are correlated (both increase in summer), but one does not cause the other.
4. **Central Limit Theorem**:
   - States that the sampling distribution of the mean will approximate a normal distribution as the sample size increases, regardless of the population's distribution.

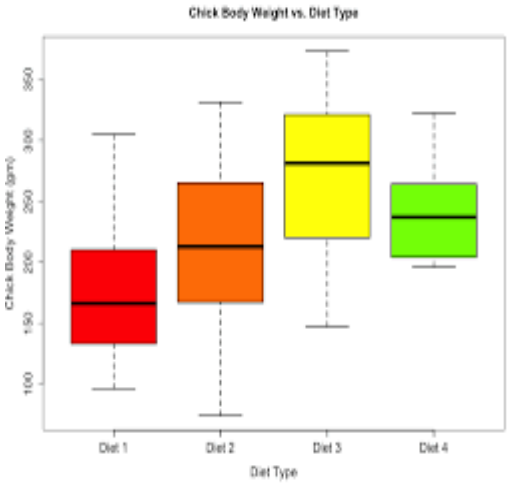
Steps in Statistical Analysis

1. **Define the Problem**: Clearly state the research question or hypothesis.
2. **Collect Data**: Gather data through surveys, experiments, or existing datasets.
3. **Clean Data**: Handle missing values, outliers, and errors.
4. **Analyze Data**: Apply descriptive and inferential statistics.
5. **Interpret Results**: Draw conclusions and make recommendations.
6. **Present Findings**: Use visualizations and reports to communicate results.

Visualization Example

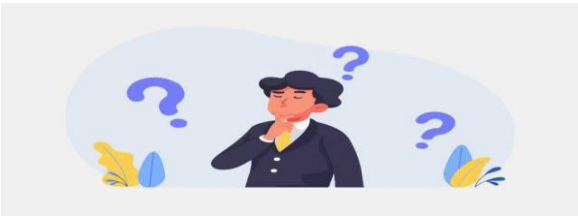






- **Histogram**: Shows the distribution of a single variable (e.g., employee salaries).
- **Scatter Plot**: Visualizes the relationship between two variables (e.g., education level vs. salary).
- **Box Plot**: Displays the median, quartiles, and outliers in a dataset.

Interview Questions



Beginner Level

1. **What is the difference between population and sample?**
    - A population is the entire set of individuals or items of interest, while a sample is a subset of the population used to make inferences about the population.
2. **What is the mean, median, and mode?**
    - Mean: The average of all values.

- Median: The middle value when data is sorted.
- Mode: The most frequently occurring value.

3. **What is the difference between variance and standard deviation?**
   - Variance measures the spread of data points around the mean, while standard deviation is the square root of variance and provides a measure of spread in the same units as the data.
4. **What is a normal distribution?**
   - A bell-shaped distribution where most of the data points cluster around the mean, with symmetrical tails on either side.
5. **What is a p-value?**
   - The p-value is the probability of observing the data (or something more extreme) if the null hypothesis is true. A low p-value (typically < 0.05) suggests rejecting the null hypothesis.

Medium Level

1. **What is the Central Limit Theorem (CLT)?**
   - The CLT states that the sampling distribution of the mean of any independent, random variable will be approximately normal if the sample size is large enough, regardless of the population's distribution.
2. **What is the difference between correlation and causation?**
   - Correlation measures the relationship between two variables, while causation implies that one variable directly affects the other. Correlation does not imply causation.
3. **What is the difference between Type I and Type II errors?**
   - Type I error: Rejecting a true null hypothesis (false positive).
   - Type II error: Failing to reject a false null hypothesis (false negative).
4. **What is the purpose of hypothesis testing?**
   - Hypothesis testing is used to make decisions or inferences about a population parameter based on sample data, using a formal statistical framework.
5. **What is the difference between a t-test and a z-test?**
   - A t-test is used when the sample size is small and the population standard deviation is unknown, while a z-test is used when the sample size is large and the population standard deviation is known.

Experienced Level

1. **What is multicollinearity, and how does it affect regression models?**
   - Multicollinearity occurs when two or more independent variables in a regression model are highly correlated. It can make it difficult to determine the individual effect of each variable and can inflate the variance of the coefficient estimates.
2. **What is the difference between parametric and non-parametric tests?**

- Parametric tests assume that the data follows a specific distribution (e.g., normal distribution), while non-parametric tests do not make such assumptions and are used for data that does not meet the assumptions of parametric tests.

3. **What is the purpose of A/B testing, and how do you analyze the results?**
   - A/B testing is used to compare two versions of a product or feature to determine which performs better. Results are analyzed using statistical tests (e.g., t-tests or chi-square tests) to determine if the differences are statistically significant.
4. **What is the difference between supervised and unsupervised learning?**
   - Supervised learning uses labeled data to train models, while unsupervised learning uses unlabeled data to find patterns or groupings in the data.
5. **How do you handle missing data in a dataset?**
   - Techniques include imputation (replacing missing values with mean, median, or mode), using algorithms that support missing data, or removing rows/columns with missing values if appropriate.

Statistics Project

1. Predictive Analysis for House Prices

- **Objective:** Predict house prices based on features like location, square footage, number of bedrooms, etc.
- **Datasets:** Kaggle: House Prices Dataset
- **Statistical Techniques:** Regression analysis, residual plots, and hypothesis testing.

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2. Customer Segmentation Using Clustering

- **Objective:** Segment customers based on purchasing behavior for targeted marketing strategies.
- **Datasets:** UCI Machine Learning Repository - Customer Data
- **Statistical Techniques:** K-Means clustering, Silhouette analysis, and Principal Component Analysis (PCA).

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3. Sentiment Analysis for Social Media Data

- **Objective:** Analyze sentiments from tweets or social media posts related to a product or topic.
- **Datasets:** Twitter API or Sentiment140 Dataset
- **Statistical Techniques:** Text preprocessing, word frequency analysis, and Chi-square tests.

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4. Sports Performance Analysis

- **Objective:** Analyze player or team performance to identify winning factors in sports.
- **Datasets:** Kaggle Sports Datasets
- **Statistical Techniques:** ANOVA, correlation analysis, and regression modeling.

5. Credit Card Fraud Detection

- **Objective:** Build a model to detect fraudulent transactions.
- **Datasets:** Kaggle Credit Card Fraud Detection Dataset
- **Statistical Techniques:** Hypothesis testing, anomaly detection, and logistic regression.

Platforms to Get statistics Questions (GitHub, LeetCode, HackerRank)

Beginner:

1. **GitHub:**
    - Search for repositories like "statistics beginner exercises" or "statistics tutorials."
    - Example: Statistics Beginner Projects
    - Questions:
        - Calculate mean, median, and mode of a dataset.
        - Compute standard deviation and variance.

Code Example (Mean, Median, Mode):

Python Code

```
import numpy as np
from scipy import stats
data = [1, 2, 2, 3, 4, 5]
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", stats.mode(data)[0][0])
```

2. **LeetCode:**
    - Search for "statistics" or "data analysis" problems.
    - Example: LeetCode Statistics Problems
    - Questions:
        - Calculate the moving average of a dataset.
        - Find the correlation between two variables.

Code Example (Moving Average):

Python Code

```
def moving_average(data, window_size):
    return np.convolve(data, np.ones(window_size), 'valid') / window_size
```

```
data = [1, 2, 3, 4, 5, 6]
print(moving_average(data, 3))
```

3. **HackerRank:**
    - Explore the "statistics" or "data science" sections.
    - Example: HackerRank Statistics Challenges
    - Questions:
        - Calculate the probability of an event.
        - Perform basic descriptive statistics.

Code Example (Probability):

Python Code

```
def probability(event_outcomes, total_outcomes):
    return event_outcomes / total_outcomes
print(probability(3, 6))  # Probability of rolling a 1, 2, or 3 on a 6-sided die
```

Medium:

1. **GitHub:**
    - Search for "statistics intermediate projects" or "statistical modeling."
    - Example: Statistics Intermediate Projects
    - Questions:
        - Perform hypothesis testing (e.g., t-test, chi-square test).
        - Fit a linear regression model.

Code Example (T-Test):

Python Code

```
from scipy import stats
group1 = [20, 22, 19, 20, 21]
group2 = [18, 19, 20, 17, 18]
t_stat, p_value = stats.ttest_ind(group1, group2)
print("T-statistic:", t_stat, "P-value:", p_value)
```

2. **LeetCode:**
    - Search for "statistics" or "data analysis" problems.
    - Example: LeetCode Statistics Problems
    - Questions:

- Implement a Monte Carlo simulation.
- Calculate the confidence interval for a dataset.

Code Example (Monte Carlo Simulation):

Python Code

```
import numpy as np
def monte_carlo_pi(n):
    inside = 0
    for _ in range(n):
        x, y = np.random.random(), np.random.random()
        if x**2 + y**2 <= 1:
            inside += 1
    return (inside / n) * 4
print(monte_carlo_pi(100000))
```

3. **HackerRank:**
   - Explore the "statistics" or "data science" sections.
   - Example: HackerRank Statistics Challenges
   - Questions:
     - Perform ANOVA (Analysis of Variance).
     - Calculate the p-value for a given hypothesis test.

Code Example (ANOVA):

Python Code

```
from scipy import stats
group1 = [20, 22, 19, 20, 21]
group2 = [18, 19, 20, 17, 18]
group3 = [22, 23, 21, 22, 20]
f_stat, p_value = stats.f_oneway(group1, group2, group3)
print("F-statistic:", f_stat, "P-value:", p_value)
```

Experienced:

1. **GitHub:**
   - Search for "statistics advanced projects" or "Bayesian statistics."
   - Example: Statistics Advanced Projects

- Questions:
  - Implement Bayesian inference.
  - Perform time series analysis (e.g., ARIMA).

Code Example (Bayesian Inference):

Python Code

```
import pymc3 as pm
with pm.Model() as model:
    p = pm.Uniform('p', 0, 1)  # Prior
    obs = pm.Binomial('obs', n=10, p=p, observed=4)  # Likelihood
    trace = pm.sample(1000)
print(pm.summary(trace))
```

2. **LeetCode:**
   - Search for "statistics" or "data analysis" problems.
   - Example: LeetCode Advanced Statistics Problems
   - Questions:
     - Implement a Markov Chain Monte Carlo (MCMC) algorithm.
     - Perform multivariate statistical analysis.

Code Example (MCMC):

Python Code

```
import numpy as np
def mcmc_sampler(target_dist, n_samples, initial_state):
    samples = [initial_state]
    for _ in range(n_samples):
        proposal = np.random.normal(samples[-1], 1)
        acceptance_ratio = target_dist(proposal) / target_dist(samples[-1])
        if np.random.random() < acceptance_ratio:
            samples.append(proposal)
        else:
            samples.append(samples[-1])
    return samples
target_dist = lambda x: np.exp(-x**2 / 2)  # Standard normal distribution
print(mcmc_sampler(target_dist, 1000, 0))
```

3. **HackerRank:**

- Explore the "statistics" or "data science" sections.
- Example: HackerRank Advanced Statistics Challenges
- Questions:
  - Perform logistic regression.
  - Calculate the Akaike Information Criterion (AIC) for a model.

Code Example (Logistic Regression):

Python Code

```
from sklearn.linear_model import LogisticRegression
X = [[1, 2], [2, 3], [3, 4], [4, 5]]
y = [0, 0, 1, 1]
model = LogisticRegression()
model.fit(X, y)
print(model.predict([[5, 6]]))
```

Summary:

- **GitHub**: Great for finding projects and code examples.
- **LeetCode**: Ideal for algorithmic statistics problems.
- **HackerRank**: Offers structured challenges with code evaluation.